NETWORK TRAFFIC ANALYSIS REPORT

Analysis Tool: Wireshark Packet Capture

Date: June 3, 2026

Host IP: 192.168.8.107

1. Executive Summary

This report presents a detailed analysis of network traffic captured via Wireshark on a local host machine (192.168.8.107). The capture reveals normal browsing behaviour with DNS resolution, TCP connection establishment, QUIC/HTTP3 encrypted sessions, and standard ARP network discovery. No immediately suspicious activity was identified; however, notable observations include QUIC handshake retransmissions and ARP broadcast patterns.

2. Network Environment

2.1 Observed Hosts & Roles

IP Address	MAC / Identity	Role	Notes
192.168.8.107	Local Host	Client Machine	Initiates all requests
192.168.8.1	Router / Gateway	Default Gateway & DNS	DNS resolver
142.251.47.46	Google Server	Remote Web Server	Google QUIC endpoint
142.251.154.119	Google Server	Remote Web Server	Resolved via DNS A record
154.0.168.115	Unknown Server	Remote Server	TCP/443 (HTTPS)
192.168.8.101	HuaweiTechno_26:e2:...	LAN Device	Subject of ARP query
192.168.8.103	HuaweiTechno_26:e2:...	LAN Device	Subject of ARP query

3. Protocol Analysis

3.1 DNS Resolution (Packets 308–309)

Standard DNS query/response flow was observed for www.google.com via the local gateway.



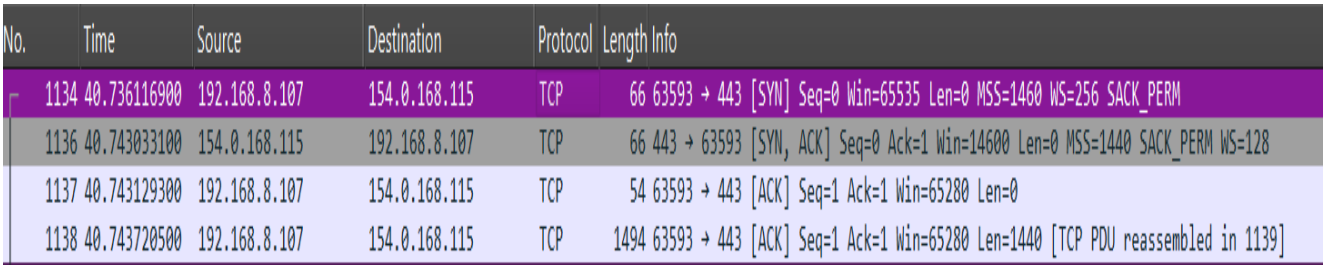
308	35.898770500	192.168.8.107	192.168.8.1	DNS	74 Standard query 0xc35f A www.google.com
309	35.900712000	192.168.8.1	192.168.8.107	DNS	202 Standard query response 0xc35f A www.google.com A 142.251.154.119 A 142

Key Observations:

- Query type A (IPv4 address record) for www.google.com with transaction ID 0xc35f.
- Response returned two A records: 142.251.154.119 and 142.251.47.46, both Google IP ranges.
- Round-trip resolution time: approximately 1.9 ms — well within normal range.
- DNS traffic routed through the local gateway (192.168.8.1) acting as a DNS forwarder.

3.2 TCP Three-Way Handshake (Packets 1134–1138)

A standard TCP connection was initiated to remote server 154.0.168.115 on port 443 (HTTPS).



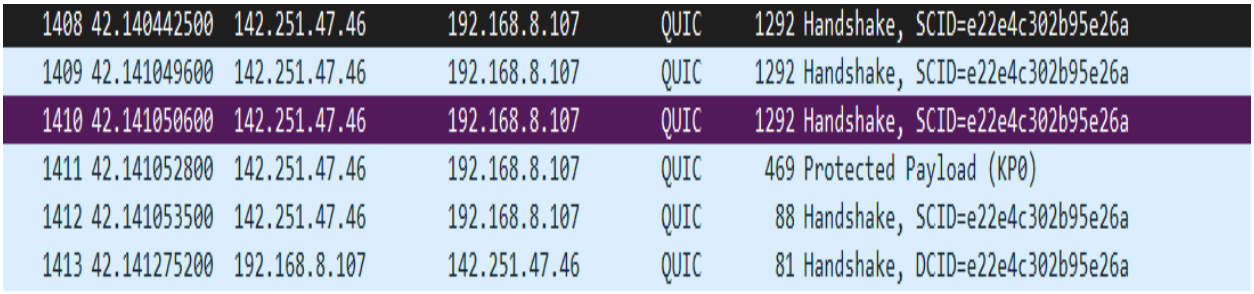
No.	Time	Source	Destination	Protocol	Length	Info
1134	40.736116900	192.168.8.107	154.0.168.115	TCP	66	63593 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1136	40.743033100	154.0.168.115	192.168.8.107	TCP	66	443 → 63593 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1440 SACK_PERM WS=128
1137	40.743129300	192.168.8.107	154.0.168.115	TCP	54	63593 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
1138	40.743720500	192.168.8.107	154.0.168.115	TCP	1494	63593 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1440 [TCP PDU reassembled in 1139]

Key Observations:

- Clean SYN → SYN-ACK → ACK handshake completed successfully.
- Client advertises Window Size 65,535 bytes; server responds with 14,600 bytes (relatively conservative).
- Both sides negotiate SACK_PERM (Selective Acknowledgement) for efficient retransmission.
- Client MSS 1460 bytes vs server MSS 1440 bytes — server uses slightly smaller segment size.
- Window Scaling enabled: client WS=256, server WS=128 — allowing large effective window sizes.
- Packet 1138 immediately follows the handshake with 1440 bytes of data, indicating immediate data transfer to port 443 (TLS/HTTPS).

3.3 QUIC / HTTP3 Traffic (Packets 1408–1413)

QUIC protocol traffic was captured communicating with Google servers (142.251.47.46), indicating HTTP/3 usage for likely Google services or Chrome browser activity.



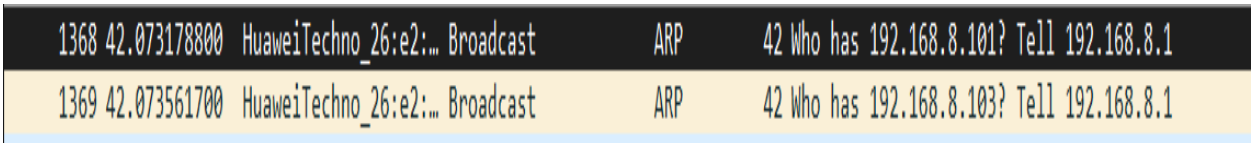
1408	42.140442500	142.251.47.46	192.168.8.107	QUIC	1292 Handshake, SCID=e22e4c302b95e26a
1409	42.141049600	142.251.47.46	192.168.8.107	QUIC	1292 Handshake, SCID=e22e4c302b95e26a
1410	42.141050600	142.251.47.46	192.168.8.107	QUIC	1292 Handshake, SCID=e22e4c302b95e26a
1411	42.141052800	142.251.47.46	192.168.8.107	QUIC	469 Protected Payload (KP0)
1412	42.141053500	142.251.47.46	192.168.8.107	QUIC	88 Handshake, SCID=e22e4c302b95e26a
1413	42.141275200	192.168.8.107	142.251.47.46	QUIC	81 Handshake, DCID=e22e4c302b95e26a

Key Observations:

- QUIC operates over UDP, providing reduced latency compared to TCP-based HTTP/2.
- Connection ID (SCID/DCID): e22e4c302b95e26a — used to maintain connection state without IP/port binding.
- Packets 1408–1410 represent handshake retransmissions (same SCID, same 1292-byte length). This may indicate mild packet loss or network jitter during the QUIC handshake phase.
- Packet 1411 carries a Protected Payload with Key Phase 0 (KP0), indicating successful 1-RTT encryption negotiation.
- Packet 1413 is the first response from the client back to Google, completing the handshake exchange.

3.4 ARP Broadcast Activity (Packets 1368–1369)

ARP (Address Resolution Protocol) broadcast requests were observed from a Huawei device on the local network.



1368	42.073178800	HuaweiTechno_26:e2:…	Broadcast	ARP	42 Who has 192.168.8.101? Tell 192.168.8.1
1369	42.073561700	HuaweiTechno_26:e2:…	Broadcast	ARP	42 Who has 192.168.8.103? Tell 192.168.8.1

Key Observations:

- ARP requests sent to Broadcast address — standard layer 2 discovery mechanism.
- The gateway (192.168.8.1) is asking the network to identify MAC addresses for .101 and .103 — likely refreshing its ARP table or resolving hosts for routing.
- Source is identified as a Huawei Technology device by MAC OUI prefix.
- No ARP replies were captured in this window — the queried hosts may be offline or slow to respond.

5. Findings

Normal browsing pattern: DNS → TCP/TLS → QUIC sequence consistent with modern browser behaviour (likely Chrome using HTTP/3 for Google services).

- QUIC handshake retransmissions (Pkts 1408–1410): Three near-identical 1292-byte handshake packets in under 1 ms suggests possible network congestion or packet drop during QUIC negotiation. Warrants monitoring.
- ARP queries with no captured replies: The gateway queried .101 and .103 but no replies appeared. These hosts may be inactive, or replies may have been filtered out of the capture scope.
- Dual Google IPs returned in DNS: Both 142.251.154.119 and 142.251.47.46 received, indicating Google CDN load-balancing DNS responses.
- Unknown server 154.0.168.115: TCP connection on port 443 was established to this IP. Its identity was not resolved in the capture. Further investigation may be warranted.

6. Conclusion

The captured network traffic reflects normal end-user browsing behaviour from host 192.168.8.107 on a home or small office LAN. Standard protocols — DNS, TCP, QUIC, and ARP — are all operating within expected parameters. The most noteworthy observations are the QUIC handshake retransmissions and an unidentified HTTPS server (154.0.168.115), both of which merit further investigation but do not on their own indicate malicious activity.